
Neoadjuvant Proton Beam Radiotherapy Combined With Subsequent Endoresection of Choroidal Melanomas

■ Nikolaos E. Bechrakis, MD

■ Michael H. Foerster, MD

The majority of patients with choroidal melanoma are currently managed by eye-retaining treatments, mainly with various irradiation modalities. Since the first successful ocular radiotherapy by Moore in the 1930s, eye-preserving treatment methods have evolved. Preference is given to ruthenium brachytherapy and proton teletherapy in Europe, whereas iodine brachytherapy and proton teletherapy are most widely applied in the United States.¹⁻⁴ The treatment of large choroidal melanomas is challenging, because radiation complications occur more frequently in that size cohort.⁵⁻⁷ The probability of radiation retinopathy and radiation-induced optic neuropathy increases with the degree of macula and optic nerve exposure and is accompanied by the sequelae of resultant tumor necrosis.⁸ A large number of eyes with irradiated large melanomas are lost as a result of neovascular glaucoma.

Surgical resection of large choroidal melanomas may have advantages over radiation monotherapy in terms of reducing both radiation complications and long-term vision loss.⁹⁻¹¹ The limitation of transscleral resection, however, lies particularly in the excision of posteriorly located tumors in close proximity to the optic disc and/or fovea, where a transscleral approach is associated with a markedly increased complication rate in large-diameter tumors and those with a large collar-button area of retinal invasion.^{12,13}

Endoresection of choroidal melanomas is an alternative surgical resection technique in which the tumor is excised by a vitreous cutter during pars plana vitrectomy providing access to tumors located posteriorly near the optic disc and/or macula.¹⁴⁻¹⁸ A potential problem with the technique is intraoperative dissemination of tumor cells that can lead to diffuse and distant recurrences.¹⁹⁻²¹ Aiming to avoid these serious

complications and drawing on long-term experience with proton teletherapy of choroidal melanomas, we developed a new neoadjuvant treatment approach applying primarily a tumoricidal proton beam irradiation and subsequently performing the endoresection of the tumor within a few days thereafter.

This study presents our 4-year experience obtained with this new combined radiation and surgical procedure.

■ Patients and Methods

After appropriate informed consent, a total of 58 consecutive patients underwent planned endoresection after proton beam irradiation as a primary treatment of choroidal melanoma from April 2000 to March 2004.

The indication for this combined procedure was for tumors whose thickness exceeded 7.0 mm and whose posterior tumor margin was located posterior to the equator, which were unsuitable for transscleral resection *ab externo* because of their posterior position. Tumors with ciliary body and/or subfoveal infiltration and tumors with a circumpapillary growth exceeding half of the optic disc circumference ($>180^\circ$) were excluded from endoresection.

The first step in the combined procedure was proton irradiation planning. For this purpose, the tumor borders (target volume) were marked by episcleral suturing of 4 or 5 tantalum clips under general anesthesia. The clip locations and tumor border positions were marked on a photomontage of the entire ocular fundus. The distance between the clips and their distance from the tumor margins, the limbus, and the optic nerve were measured intraoperatively and postoperatively by ultrasound and computed tomography.

Model-based irradiation planning was done with the program EYEPLAN.²² Technical details regarding the radiation unit and the positioning procedure have already been described elsewhere.^{23,24}

The endoresection was performed between 4 and 21 days (median, 9 days) after proton therapy by either author. Five patients with highly prominent tumors who had either preexistent cataracts or an anterior tumor margin near the posterior lens pole underwent phacoemulsification with implantation of a foldable posterior chamber lens before vitrectomy in the same session. During 3-port pars plana vitrectomy, a complete posterior vitreous detachment was induced by suction with the vitreous cutter, care being taken to also detach the vitreous body from the anterior tumor margin. After endodiathermy of the retinal vessels over the tumor base, the subretinal fluid was drained through a retinectomy with application of perfluorodecalin. This was followed by transretinal endoresection with increased intraocular pressure (up to 85 mm Hg) and low systolic

blood pressure (≤ 90 mm Hg) to minimize bleeding from tumor vessels. Aspiration pressure was set to 200 mm Hg and cut rate to 600 cuts/min. The tumor was excised up to the sclera accompanied by endodiathermy of individual scleral ciliary arteries and choroidal vessels. After flattening the retina by further perfluorodecalin injection, the retinectomy and choroidectomy edges were coagulated by an endolaser in continuous-wave mode and by cryocoagulation for the anterior endoresection and retinectomy borders. Flat tumor residues still remaining at the bare sclera were subjected to hyperthermia in the sense of “endothermic therapy” by the use of an 810-nm diode laser. Perfluorodecalin was then directly exchanged for silicone or, in 4 cases, replaced by an air-SF₆ (30%) injection.

Patients were followed up at regular 3-month intervals, and silicone removal was usually performed after approximately 4 months in combination with phacoemulsification and implantation of a foldable posterior chamber lens (Fig. 5, Fig. 6).

Statistical Analysis

The statistical analysis included medians, deviations, and ranges for continuous variables as well as absolute and relative frequencies for dichotomous variables. The Kaplan-Meier (KM) method was used to evaluate time-to-event data. Risk factors for visual loss were analyzed with multivariate analysis (linear regression and Cox proportional hazards model). To simplify the statistical analysis, the measured decimal vision was converted into logMAR (negative logarithm of the minimal angle of resolution) units and converted after statistical analysis back to decimal vision.²⁵

■ Results

A total of 58 consecutive patients underwent endoresection after proton teletherapy with 60 Gy applied in 4 fractions. Table 1 summarizes patient and tumor data. The median preoperative vision was 20/40 and varied between 20/20 and 20/400. One year after endoresection, the median visual acuity was reduced to 20/200 and stayed stable at this level throughout the follow-up period (Fig. 4).

In 21 patients, a cataract operation was either combined with the silicone removal ($n = 19$) or performed alone ($n = 2$). The probability of developing a cataract within the first 2 postoperative years was estimated at 46.6% using the KM method.

In the postoperative phase, 16 of 58 patients (28%) developed a retinal detachment that was treated by repeat vitrectomy, buckling procedures, or both. The probability of retinal detachment within the first 24 postoperative months was estimated at 32.1% using the KM method (Table 2).

Table 1. *Patient and Tumor Data*

	Median	Mean	Min.	Max.	
Age	55.8	56.9	15.7	82.6	(years)
Tumor thickness	8.8	9.1	7.0	14.5	(mm)
Largest tumor diameter	15.6	15.4	11.1	21.6	(mm)
Distance PTM — optic disc	2.2	2.7	0.0	11.0	(mm)
Distance PTM — fovea	3.0	3.2	0.0	8.1	(mm)
Follow-up time	18.0	21.7	3.0	48.0	(months)
Visual acuity	20/40	20/50	20/400	20/20	

PTM: posterior tumor margin.

Silicone was removed after a median period of 4 months. One patient had to have revision surgery because of a phakic silicone prolapse in the anterior chamber and significant bleeding within the tumor bed on the first postoperative day. A macular hole was found in 2 patients 1 and 12 months after endoresection and a macular pucker in another patient 6 months thereafter (Table 2).

Three patients underwent enucleation in conjunction with no light perception of the affected eye as a result of phthisis, secondary glaucoma, or complicated postvitrectomy retinal detachments. The KM estimate for eye retention 24 months after endoresection was 91.6% (Table 3; Fig. 1). One further eye had to be enucleated 40 months after endoresection as a result of tumor recurrence.

Radiation retinopathy was found in a total of 12 patients and radiation optic neuropathy in a total of 13 patients. The probability of developing radiation retinopathy or radiation optic neuropathy after 24 months was estimated at 27.5% and 29.4% with the KM method (Table 3; Fig. 2 and 3). The earliest signs of radiation retinopathy or neuropathy were seen 6 months after surgery in one patient who had a juxtapapillary melanoma with a tumor thickness of 7.0 mm and a maximal diameter of 18.3 mm. This patient was also the only one in this treatment series who developed

Table 2. *Kaplan Meier Estimates for Surgical Complications After 24 Months*

	n	%
Cataract	21	46.6%
Retinal detachment	16	32.1%
Macular hole	2	4.0%
Secondary glaucoma	1	2.6%
Phthisis	1	2.1%
Macular pucker	1	1.8%

Table 3. *Kaplan Meier Estimates for Tumor Related Complications After 24 Months*

	n	%
Enucleation*	3	8.4%
Radiation retinopathy**	11	27.5%
Radiation optic neuropathy	13	29.4%
Local tumor recurrence [†]	0	0.0%
Metastases [‡]	2	4.0%

*A fourth enucleation was performed 40 months after endoresection.

**A twelfth radiation retinopathy occurred 36 months after endoresection.

[†]One local tumor recurrence occurred 40 months after endoresection.

[‡]A third metastasis occurred 27 months after endoresection.

neovascularization glaucoma 18 months after surgery although he had undergone panretinal laser coagulation 12 months after endoresection. No evidence of tumor recurrence was found after histopathologic examination of the 3 enucleated eyes as a result of phthisis, secondary glaucoma, or complicated retinal detachment. A fourth enucleation was necessary 40 months after endoresection as a result of marginal tumor recurrence as was confirmed after histopathologic examination. Liver metastases were found in 3 patients 1 to 27 months after endoresection (Table 3).

All possible risk factors for visual loss were included in a multivariate linear regression model. The only significant risk factor after forward and backward selection was largest tumor diameter ($P = 0.001$).

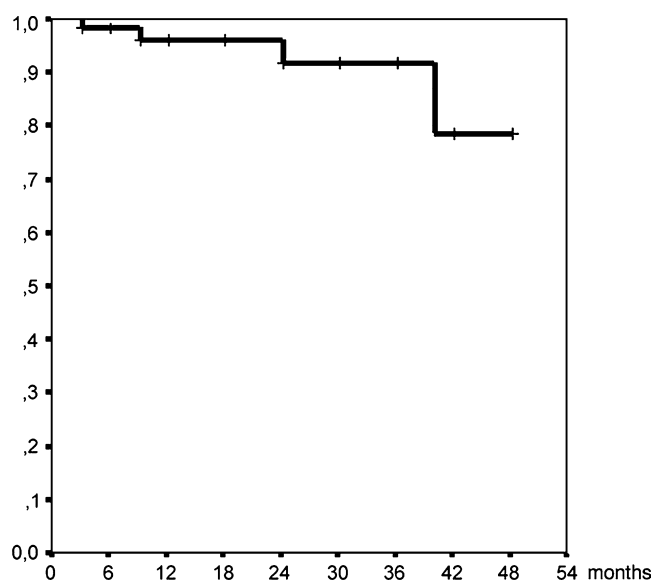


Figure 1. *Estimated Kaplan-Meier "eye retention" curve. The curve is based on the probability of avoiding enucleation after endoresection.*

■ Discussion

Radiotherapy is the mainstay of choroidal melanoma treatment, and many authors have reported very good local tumor control rates with various irradiation modalities.^{6,7,26,27} Depending on the treatment period and institution, local tumor control rates are 70% to 89% for ruthenium brachytherapy,^{6,28} 81% to 96% for iodine brachytherapy,^{3,6,29,30} and 91% to 99% for proton teletherapy.^{6,26,27} Retrospective analyses have not revealed any differences in mortality between the different treatment modalities. The Collaborative Ocular Melanoma Study (COMS), which has been in progress for 20 years, likewise showed no difference in the survival prognosis between enucleation and eye-preserving iodine brachytherapy.³

There is consensus that certain tumor parameters significantly influence both mortality and morbidity. It is accepted that mortality cannot yet be influenced by modifying and optimizing eye-preserving treatment modalities. This is not the case with respect to ocular morbidity, however; advances have been achieved by the development of new irradiation strategies and surgical techniques. The most important prognostic factor regarding ocular morbidity is the size and location of the tumor.^{8,24,31} It has long been known that the probability of developing vision-threatening complications increases with the size of the tumor and the proximity of its borders to the optic disc and macula.^{12,32-34} This is not only the result of increased radiation exposure of these vulnerable structures, which can cause late complications like radiation retinopathy or radiation optic

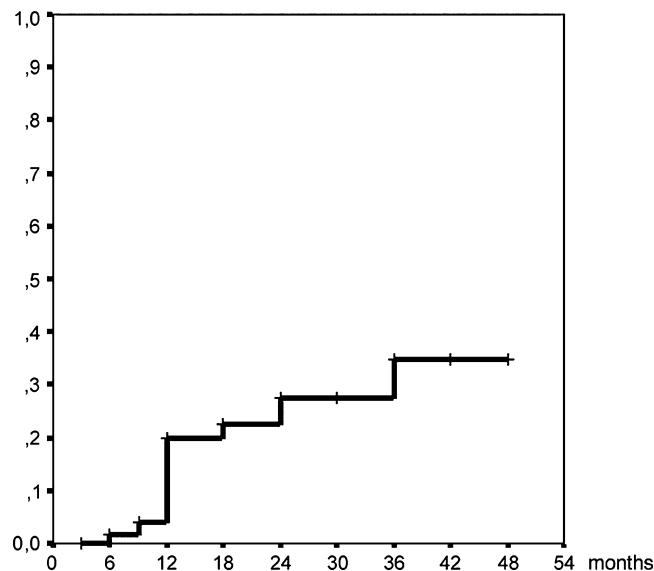


Figure 2. Kaplan-Meier curve showing the estimated probability of developing radiation retinopathy.

neuropathy, but also to an early postirradiation inflammatory reaction that apparently depends on the extent of induced tumor necrosis as well as on the acute tumor vascular damage. An enlargement of the treated tumor volume is accompanied by increased induction of inflammatory factors, including the production of cytokines like VEGF, $\text{TNF}\alpha$, and so on, and the accumulation of tumor-reducing macrophages and other direct tumor-degradation products. This promotes the development of exudative retinal detachment and secondary glaucoma such as the neovascular or melanomalytic type.³⁵ These problems can be largely eliminated by surgical removal of the tumor, like in transscleral resection (ab externo). The major limiting factor of transscleral resection ab externo is, however, the increasing operative difficulty and complication rate associated with tumors whose posterior margin is situated near the optic disc and/or macula.¹²

Internal eye wall resection alias endoresection was described by Peyman and then further developed by Damato as approaches for removing tumors near the optic disc or macula.^{14–17} These procedures were performed either primarily without any preceding therapy intending to avoid radiogenic side effects to the macula and/or the optic disc or as so-called secondary “salvage procedures” after various preceding therapies aiming at reducing the sequelae of the primary treatment such as retinal detachment and secondary glaucoma. However, these surgical techniques have been justly criticized because of the severe risk of tumor recurrences. Merely ablating the tumor with a vitreous cutter could not ensure that it

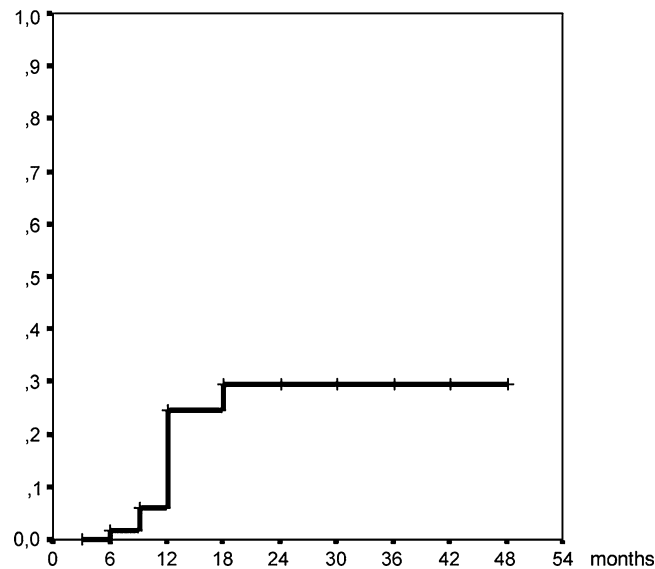


Figure 3. Kaplan-Meier curve showing the estimated probability of developing radiation optic neuropathy.

was completely removed and that no vital tumor cells were spread by the vitrectomy to the different tissue compartments. These risks have recently been confirmed by the description of intra- and extraocular recurrences both close to and distant from the tumor.^{19–21}

Intraocular tumor recurrence has a negative influence on patients' survival prognosis regardless of the primary treatment modality. This was in part clearly evident in most retrospective analyses, and even the prospective COMS has shown that tumor recurrences tend to worsen the prognosis.^{3,27,36–38} It is now generally accepted that adjuvant radiation after transscleral tumor resection can reduce the incidence of both recurrences and metastases.^{13,39,40} This surgical technique enables excision of the tumor in one piece in the sense of a lamellar sclerochoroidectomy. Because the sites most susceptible to tumor recurrences are the potentially incompletely excised tumor margins, ruthenium brachytherapy or proton therapy is performed immediately after tumor resection. Thus, the resection margins at the scleral level can be submitted to an effective scleral contact dose of adjuvant postoperative irradiation without endangering critical structures or the eye as a whole. The principle of adjuvant postoperative irradiation is not transferable to a planned endoresection (ab interno). With this surgical technique, the tumor is gradually ablated layer by layer from the interior part of the eye with a vitreous cutter. During surgery, tumor cells can be potentially disseminated throughout the eye as well as through the sclerotomies into the episcleral space. Moreover, tumor cells could be inserted into the general circulation through the incised vessels. In such a situation, postoperative

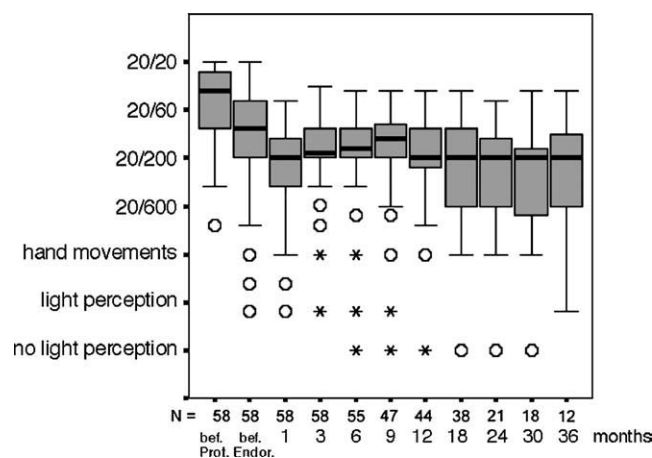


Figure 4. “Box and whisker” plot of the development of visual acuity after endoresection. The gray “boxes” represent the central 50% of the data and are further divided by the thick horizontal line, which is the median value. The “whiskers” represent the minimum and maximum values. Outlying observations are marked with small circles and asterisks.

irradiation with any modality in the tumor bed would not be useful, because vital tumor cells could have been spread to areas outside the irradiation field. In this surgical setting, only a preoperative irradiation that covers and “inactivates” all tumor cells can be protective with respect to local or distant tumor recurrence.

The optimal irradiation modality would be characterized by a homogeneous dose distribution within the target volume and a steep dose reduction outside the target volume to protect adjacent radiosensitive structures. These demands are fully met by the physical properties of the proton beam, whose dose increases with the penetration depth and then shows an extremely steep cutoff in the so-called “Bragg peak.”⁴¹ Two recent reports have published preliminary data of endoresection after radiation pretreatment either by proton beam irradiation or by applying stereotactic



Figure 5. Photomontage of a posteriorly located choroidal melanoma. The tumor thickness is 8.3 mm and the visual acuity 20/50.

single-dose convergence irradiation with a Leksell gamma knife and additionally performed ruthenium brachytherapy after endoresection if the tumor could not be completely excised.^{42,43}

We report on our experience and longer follow up of this new neoadjuvant combined radiotherapeutic–surgical approach, in which 4 fractions of proton irradiation with a total of 60 CGE are applied a few days before performing planned transretinal endoresection. Because 60 CGE of proton beam irradiation is regarded as a curative dose,²⁴ endoresection was still performed with the aim of complete tumor removal, but a surgical safety margin was dispensed to avoid excising vision-relevant structures near the optic disc or macula. It is still unclear at present whether a curative radiation dose has to be applied or whether a reduced dose would also be sufficient for preventing recurrences. In a prospective study, Gragoudas reported that a reduced dose of 50 CGE instead of the 70 CGE did not increase significantly the recurrence rate, but it also failed to reduce the

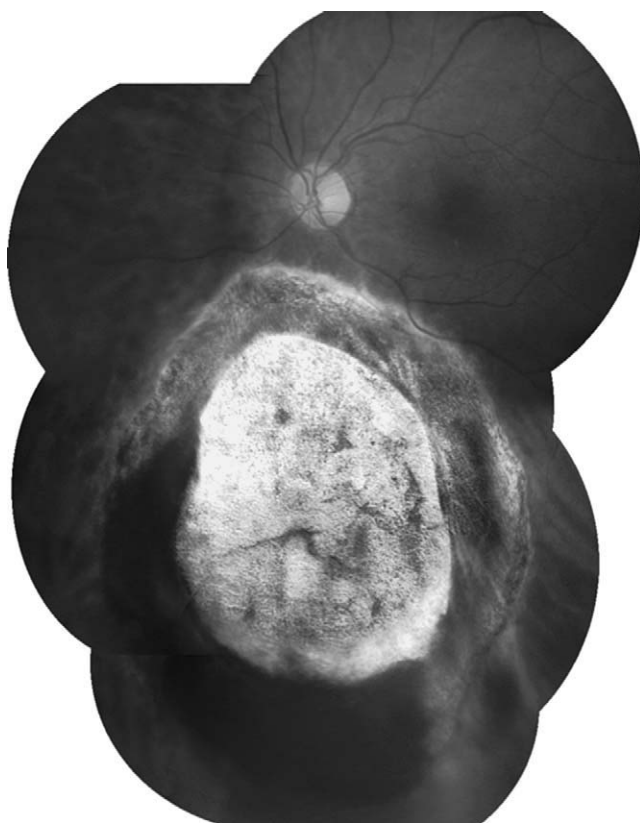


Figure 6. Photomontage of the same case as depicted in Figure 5 2 years after proton beam irradiation with subsequent endoresection. There are no signs of radiation retinopathy or radiation optic neuropathy and visual acuity is 20/100.

incidence of radiation-induced complications.⁴⁴ In our patient population, 27% to 29% of patients showed early signs of radiation retinopathy or optic neuropathy in the first 24 months after endoresection. They occurred at the earliest (in one patient) after 6 months and at a median of 12 months after irradiation. It is unclear whether the incidence of radiation complications is reduced or left unchanged by removing the tumor from the eye through endoresection. However, taking into account that all the patients treated in this study had large posteriorly located tumors, which are at the highest risk of developing radiogenic side effects, the results presented here suggest a favorable outcome in terms of reducing ocular morbidity and preserving vision, although longer follow up is needed.

An additional open question is the optimal postirradiation time window for performing surgery. Theoretically, an early operation would be advisable after radiotherapy for preventing the induction of acute or subacute inflammatory reactions. In our experience, performing the operation a few days to 2 weeks after irradiation has no negative influence on either the technical performance of surgical procedures or the postoperative morphologic and functional results.

In conclusion, endoresection after preoperative neoadjuvant proton beam irradiation is a promising procedure for treating very thick choroidal melanomas located near the posterior pole in close proximity to the optic nerve or macula. These posteriorly located tumors can only be managed by transscleral resection with increased surgical risks. Long-term observations will have to demonstrate whether this newly developed combined radiotherapeutic–surgical concept can effectively prevent serious radiation-induced eye-endangering side effects such as those associated with radiation monotherapy of large melanomas. In short-term follow up (with a mean of 2 years), these high-risk patients, in which enucleation would have been the alternative treatment option, did not have increased morbidity compared with patients with low-risk choroidal melanomas regarding treatment-related ocular side effects.

■ References

1. Lommatzsch PK, Werschnik C, Schuster E. Long-term follow-up of Ru-106/Rh-106 brachytherapy for posterior uveal melanoma. *Graefes Arch Clin Exp Ophthalmol*. 2000; 238:129–137.
2. Egger E, Zografos L, Schalenbourg A, et al. Eye retention after proton beam radiotherapy for uveal melanoma. *Int J Radiat Oncol Biol Phys*. 2003;55:867–880.
3. Jampol LM, Moy CS, Murray TG, et al. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 years after brachytherapy. COMS report no. 19. *Ophthalmology*. 2002;109: 2197–2206.
4. Gragoudas ES. 1996. Jules Gonin Lecture of the Retina Research Foundation. Long-term results after proton irradiation of uveal melanomas. *Graefes Arch Clin Exp Ophthalmol*. 1997;235:265–267.

5. Foss AJ, Whelehan I, Hungerford JL, et al. Predictive factors for the development of rubeosis following proton beam radiotherapy for uveal melanoma. *Br J Ophthalmol*. 1997;81:748–754.
6. Wilson MW, Hungerford JL. Comparison of episcleral plaque and proton beam radiation therapy for the treatment of choroidal melanoma. *Ophthalmology*. 1999;106:1579–1587.
7. Shields CL, Naseripour M, Cater J, et al. Plaque radiotherapy for large posterior uveal melanomas (> or = 8-mm thick) in 354 consecutive patients. *Ophthalmology*. 2002;109:1838–1849.
8. Gragoudas ES, Li W, Lane AM, et al. Risk factors for radiation maculopathy and papillopathy after intraocular irradiation. *Ophthalmology*. 1999;106:1571–1577; discussion 77–78.
9. Augsburger JJ, Lauritzen K, Gamel JW, et al. Matched group study of surgical resection versus cobalt-60 plaque radiotherapy for primary choroidal or ciliary body melanoma. *Ophthalmic Surg*. 1990;21:682–688.
10. Bechrakis NE, Bornfeld N, Zoller I, et al. Iodine 125 plaque brachytherapy versus transscleral tumor resection in the treatment of large uveal melanomas. *Ophthalmology*. 2002;109:1855–1861.
11. Kivela T, Puusaari I, Damato B. Transscleral resection versus iodine brachytherapy for choroidal malignant melanomas 6 millimeters or more in thickness: a matched case-control study. *Ophthalmology*. 2003;110:2235–2244.
12. Damato BE, Paul J, Foulds WS. Predictive factors of visual outcome after local resection of choroidal melanoma. *Br J Ophthalmol*. 1993;77:616–623.
13. Bechrakis NE, Bornfeld N, Zoller I, et al. Iodine-125 brachytherapy and trans-scleral tumor resection in large uveal melanomas. *Ophthalmologie*. 2001;98:736–742.
14. Peyman GA, Cohen SB. Ab interno resection of uveal melanoma. *Int Ophthalmol*. 1986;9:29–36.
15. Peyman GA, Charles H. Internal eye wall resection in the management of uveal melanoma. *Can J Ophthalmol*. 1988;23:218–223.
16. Damato B, Groenewald C, McGalliard J, et al. Endoresection of choroidal melanoma. *Br J Ophthalmol*. 1998;82:213–218.
17. Kertes PJ, Johnson JC, Peyman GA. Internal resection of posterior uveal melanomas. *Br J Ophthalmol*. 1998;82:1147–1153.
18. Garcia-Arumi J, Sararols L, Martinez V, et al. Vitreoretinal surgery and endoresection in high posterior choroidal melanomas. *Retina*. 2001;21:445–452.
19. Damato B, Wong D, Green FD, et al. Intrasceral recurrence of uveal melanoma after transretinal 'endoresection.' *Br J Ophthalmol*. 2001;85:114–115.
20. Hadden PW, Hiscott PS, Damato BE. Histopathology of eyes enucleated after endoresection of choroidal melanoma. *Ophthalmology*. 2004;111:154–160.
21. Mittica N, Vemuganti GK, Duffy M, et al. Late orbital recurrence of a choroidal melanoma following internal resection. Report of a case and review of the literature. *Surv Ophthalmol*. 2003;48:181–190.
22. Goitein M, Miller T. Planning proton therapy of the eye. *Med Phys*. 1983;10:275–283.
23. Heufelder J, Cordini D, Fuchs H, et al. [Five years of proton therapy of eye neoplasms at the Hahn-Meitner Institute, Berlin.] *Z Med Phys*. 2004;14:64–71.
24. Hocht S, Bechrakis NE, Nausner M, et al. Proton therapy of uveal melanomas in berlin 5 years of experience at the Hahn-Meitner Institute. *Strahlenther Onkol*. 2004;180:419–424.
25. Ferris FL 3rd, Kassoff A, Bailey I. New visual acuity charts for clinical research. *Am J Ophthalmol*. 1982;94:91–98.
26. Gragoudas ES, Egan KM, Seddon JM, et al. Intraocular recurrence of uveal melanoma after proton beam irradiation. *Ophthalmology*. 1992;99:760–766.
27. Egger E, Schalenbourg A, Zografos L, et al. Maximizing local tumor control and survival after proton beam radiotherapy of uveal melanoma. *Int J Radiat Oncol Biol Phys*. 2001;51:138–147.

28. Lommatzsch PK. Results after beta-irradiation ($^{106}\text{Ru}/^{106}\text{Rh}$) of choroidal melanomas: 20 years' experience. *Br J Ophthalmol*. 1986;70:844–851.
29. Quivey JM, Augsburger J, Snelling L, et al. ^{125}I plaque therapy for uveal melanoma. Analysis of the impact of time and dose factors on local control. *Cancer*. 1996;77:2356–2362.
30. Char DH, Kroll S, Phillips TL, et al. Late radiation failures after iodine 125 brachytherapy for uveal melanoma compared with charged-particle (proton or helium ion) therapy. *Ophthalmology*. 2002;109:1850–1854.
31. Char DH, Kroll S, Quivey JM, et al. Long term visual outcome of radiated uveal melanomas in eyes eligible for randomisation to enucleation versus brachytherapy. *Br J Ophthalmol*. 1996;80:117–124.
32. Egan KM, Gragoudas ES, Seddon JM, et al. The risk of enucleation after proton beam irradiation of uveal melanoma. *Ophthalmology*. 1989;96:1377–1382.
33. Seddon JM, Gragoudas ES, Polivogianis L, et al. Visual outcome after proton beam irradiation of uveal melanoma. *Ophthalmology*. 1986;93:666–674.
34. Shields CL, Shields JA, Cater J, et al. Plaque radiotherapy for uveal melanoma: long-term visual outcome in 1106 consecutive patients. *Arch Ophthalmol*. 2000;118:1219–1228.
35. Lumbroso L, Desjardins L, Levy C, et al. Intraocular inflammation after proton beam irradiation for uveal melanoma. *Br J Ophthalmol*. 2001;85:1305–1308.
36. Vrabec TR, Augsburger JJ, Gamel JW, et al. Impact of local tumor relapse on patient survival after cobalt 60 plaque radiotherapy. *Ophthalmology*. 1991;98:984–988.
37. Harbour JW, Char DH, Kroll S, et al. Metastatic risk for distinct patterns of post irradiation local recurrence of posterior uveal melanoma. *Ophthalmology*. 1997;104:1785–1792.
38. Egan KM, Ryan LM, Gragoudas ES. Survival implications of enucleation after definitive radiotherapy for choroidal melanoma: an example of regression on time-dependent covariates. *Arch Ophthalmol*. 1998;116:366–370.
39. Damato BE, Paul J, Foulds WS. Risk factors for residual and recurrent uveal melanoma after trans-scleral local resection. *Br J Ophthalmol*. 1996;80:102–108.
40. Damato BE, Paul J, Foulds WS. Risk factors for metastatic uveal melanoma after trans-scleral local resection. *Br J Ophthalmol*. 1996;80:109–116.
41. Bechrakis NE, Foerster MH. Where is the superiority of proton radiation for ocular tumours? *Graefes Arch Clin Exp Ophthalmol*. 2002;240:513–514.
42. Bechrakis N, Höcht S, Martus P, et al. Endoresektion von großen uvealen Melanomen nach Protonenbestrahlung. *Ophthalmologie*. 2004;101:370–376.
43. Bornfeld N, Talies S, Anastassiou G, et al. Endoresection of malignant melanomas of the uvea after preoperative stereotactic single dose convergence irradiation with the Leksell gamma knife. *Ophthalmologie*. 2002;99:338–344.
44. Gragoudas ES, Lane AM, Regan S, et al. A randomized controlled trial of varying radiation doses in the treatment of choroidal melanoma. *Arch Ophthalmol*. 2000;118:773–778.